CSC14: Discrete Structures for Computer Science I

1 Definition of Induction

Induction is a mathematical proof technique used to prove a statement $P(n)$ for all natural numbers n. The process consists of two main steps: the base case and the inductive step.

2 Steps to Validate a Proof by Induction

To validate a proof by induction, follow these steps:

1. **Establish the base case:** Verify that the statement holds for the initial value, typically $n = 1$.
2. **Introduce the induction hypothesis:** Assume the statement holds for some integer k, i.e., assume $P(k)$ is true.
3. **Assert** and **prove** $P(k+1)$**:** Show that under the assumption that $P(k)$ is true, the statement $P(k+1)$ is also true.

For instance, when working with positive integers, the inductive step can be viewed as stating that if $P(1)$ is true, and for all k, $P(k)$ implies $P(k+1)$, then this implies $P(x)$ for all positive integers x. In otehr words, using the logic statements we have seen so far, we can write the inductive step as:

$$(P(1) \wedge \forall k \in \mathbb{N}, P(k) \implies P(k+1)) \implies \forall x \in \mathbb{N}, P(x).$$

3 Examples

3.1 Example 1: Proving the sum of the first n positive integers

We aim to prove that

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

for all positive integers n. This means that we are trying to show that

$$1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}.$$

This material heavily borrows from the Fall 2024 syllabus for CSC14, Discrete Structures for Computer Science I, taught by Prof. Scott M. Jeffreys at Hofstra University.

Proof.

Base case ($n = 1$):

$$\sum_{i=1}^{1} i = 1 = \frac{1(1+1)}{2}.$$

Thus, the base case holds.

Inductive step: Assume the statement is true for $n = k$, i.e.,

$$\sum_{i=1}^{k} i = \frac{k(k+1)}{2}.$$

We need to show it holds for $n = k + 1$:

$$\sum_{i=1}^{k+1} i = \sum_{i=1}^{k} i + (k+1) = \frac{k(k+1)}{2} + (k+1).$$

Factoring out $(k + 1)$:

$$= \frac{k(k+1)}{2} + \frac{2(k+1)}{2} = \frac{k(k+1) + 2(k+1)}{2} = \frac{k^2 + 3k + 2}{2} = \frac{(k+1)(k+2)}{2}.$$

Hence, the statement holds for $n = k + 1$. □

By the principle of induction, the statement is true for all positive integers n.

3.2 Example 2: Proving the sum of the first n odd integers

We aim to prove that

$$\sum_{i=1}^{n} (2i - 1) = n^2$$

for all positive integers n.

Proof.

Base case ($n = 1$):

$$\sum_{i=1}^{1} (2i - 1) = 1 = 1^2.$$

Thus, the base case holds.

Inductive step: Assume the statement is true for $n = k$:

$$\sum_{i=1}^{k} (2i - 1) = k^2.$$

We need to show it holds for $n = k + 1$:

$$\sum_{i=1}^{k+1} (2i - 1) = \sum_{i=1}^{k} (2i - 1) + (2(k+1) - 1).$$

Using the induction hypothesis:

$$= k^2 + (2k + 1) = k^2 + 2k + 1 = (k + 1)^2.$$

□

Thus, the statement holds for $n = k + 1$.
By the principle of induction, the statement is true for all positive integers n.

3.3 Example 3: Proving $n < 2^n$ for all positive integers n

We aim to prove that $n < 2^n$ for all positive integers n.

Proof.

Base case ($n = 1$):

$$1 < 2^1 = 2.$$

Thus, the base case holds.

Inductive step: Assume the statement is true for $n = k$:

$$k < 2^k.$$

We need to show it holds for $n = k + 1$:

$$k + 1 < 2^k + 1 < 2 \cdot 2^k = 2^{k+1}.$$

Hence, the statement holds for $n = k + 1$. □

By the principle of induction, the statement is true for all positive integers n.

3.4 Example 4: Proving 2 divides $x^2 + x$ if x is a positive integer

We aim to prove that 2 divides $x^2 + x$ for all positive integers x.

Proof.

Base case ($x = 1$):

$$1^2 + 1 = 2,$$

which is divisible by 2.

Inductive step: Assume the statement is true for $x = k$:

$$k^2 + k \text{ is divisible by } 2.$$

We need to show it holds for $x = k + 1$:

$$(k + 1)^2 + (k + 1) = k^2 + 2k + 1 + k + 1 = k^2 + k + 2k + 2.$$

By the induction hypothesis, $k^2 + k$ is divisible by 2, and $2k + 2$ is also divisible by 2.
Thus, $(k + 1)^2 + (k + 1)$ is divisible by 2.
By the principle of induction, the statement is true for all positive integers x.

□